Module 1: Overview

MGMT 675: Generative AI for Finance

Kerry Back

From Makers to Checkers

Derek Waldron, JP Morgan Chief Analytics Officer

What we're working towards is that every employee will have their own personalized AI assistant; every process is powered by AI agents, and every client experience has an AI concierge.

You'll still have people at the top who are managing and have relationships with clients, but many, many of the processes underneath are now being done by AI systems.

Workers would shift from being creators of reports or software updates, or 'makers' ... to 'checkers' or managers of AI agents doing that work.

CNBC interview, September 30, 2025.

CFOs Are Going All In on AI

Deloitte Q4 2025 CFO Signals Survey — 200 CFOs at $1B+ companies

- **87%** say AI will be extremely or very important to finance operations in 2026
- **54%** prioritize integrating AI agents in their finance departments
- **50%** cite digital transformation of finance as their #1 priority
- **49%** prioritize automating processes to free employees for higher-value work
- Only **2%** say AI won't be important

Case Study: HPE's "Alfred"

Hewlett Packard Enterprise (#145 on Fortune 500)

The Problem

- Weekly 90-minute operational review required 100+ slides
- Hundreds of hours of preparation across business units
- No time left for forward-looking analysis

The AI Solution

- Built "Alfred"—AI agents that pull, reconcile, and analyze data automatically
- **90%** of manual prep eliminated; cycle time reduced **40%**, costs down **25%**
- 3,000+ finance employees being reskilled to build their own agents

"The goal is for finance professionals to become **masters of their own destiny** rather than casualties of automation." — Marie Myers

"How HPE's CFO used AI to transform the 100-slide Monday meeting her team spent all week preparing for," *Fortune*, Feb. 12, 2026.

Something Big is Happening

Matt Shumer, CEO of HyperWrite AI, Feb. 2025

The person who walks into a meeting and says "I used AI to do this analysis in an hour instead of three days" is going to be the most valuable person in the room. Not eventually. Right now.

Here's a simple commitment that will put you ahead of almost everyone: spend one hour a day experimenting with AI ... try to get it to do something new, something you're not sure it can handle. If you do this for the next six months, you will understand what's coming better than 99% of the people around you.

AI Agents

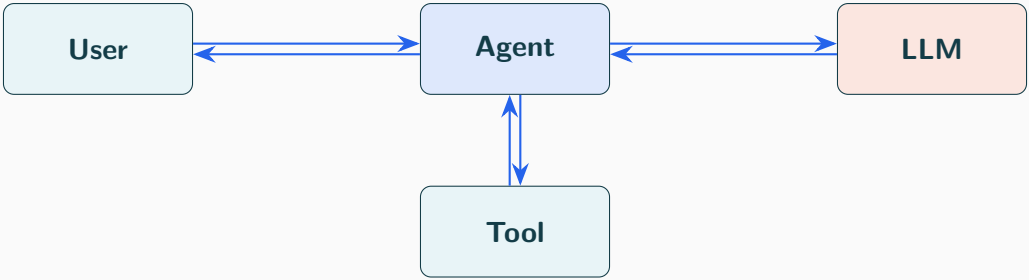
Chatbot

The chatbot sends three things to the LLM with each request:

1. **User prompt** — what you just typed
2. **System prompt** — hidden instructions that define the LLM's behavior, e.g., *"You are a helpful assistant . . . "*
3. **Conversation history** — all prior messages in the session, so the LLM has context

Anthropic publishes the system prompt for Claude:
docs.anthropic.com/en/docs/about-claude/soul

Agent



The agent programmatically controls communication between the LLM and the tool. The system prompt includes *"You have a tool . . . "* so the LLM knows what it can call. Multiple rounds of communication may occur between the LLM, agent, and tool before the agent returns output to the user.

The Agent Loop

Every agent follows a loop: **plan, execute, observe, iterate.**

1. **Plan** — the LLM reads your request and breaks it into steps
2. **Execute** — the agent calls a tool (runs code, queries a database, writes a file)
3. **Observe** — the agent checks the result of the action
4. **Iterate** — if the result is not satisfactory, the agent adjusts its plan and tries again

When an agent encounters an error, it does not stop and ask you what to do. It diagnoses the problem, revises its approach, and tries again. This **autonomous problem-solving** is what separates agents from simple automation scripts.

How Tool Calls Work

The LLM does not run code or call databases directly. When it decides a tool is needed, it returns a structured message:

```
{``tool'': ``sql_query'', ``query'': ``SELECT ...''}
```

The agent's execution layer reads that message, runs the actual tool, and sends the result back to the LLM. The LLM reads the result and decides what to do next.

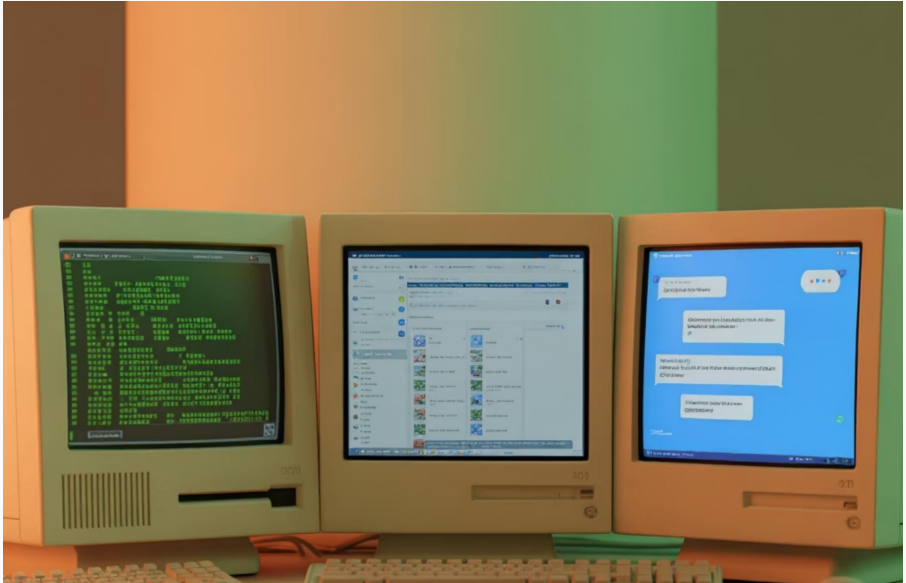
An agent can use **multiple tools in sequence**. For example: query a database, pass the results to Python for analysis, generate a chart, and assemble everything into a PowerPoint deck — all from a single instruction.

Examples of Agent Tools

- Python environment
- Database server
- Web browser / web search
- File system (read, write, edit)
- API calls (REST endpoints, data providers)

An agent may be connected to **multiple tools**. For example, get data from a database server and send it to a Python environment to return analysis.

User Interfaces: Terminal → Point and Click → Chat



Example: Rice Data Portal

The Rice Data Portal is an agent.

- **LLM:** ChatGPT — interprets your natural-language question and writes a SQL query
- **Tool:** A database server in the cloud holding stock-market and financial data
- **Agent:** An app running elsewhere in the cloud that coordinates communication between ChatGPT and the database

You type a question (*"Show me AAPL closing prices for 2024"*). The agent sends it to ChatGPT, which returns a SQL query. The agent runs the query against the database and displays the answer. If the query fails, ChatGPT rewrites it and tries again — multiple rounds, no human intervention.